

Practical Viva Important Questions

Unit 1:

1. What is the difference between HTML and HTML5?
2. What is the purpose of the <meta> tag?
3. What is the difference between id and class in CSS?
4. What is the CSS Box Model?
5. What is the difference between position: relative, absolute, fixed, and sticky?
6. What is event bubbling and event capturing in JavaScript?
7. What is the DOM in JavaScript?
8. What is AJAX and why is it used?
9. What is the difference between GET and POST in AJAX?
10. What is the difference between let, var, and const?
11. How do you center a div both horizontally and vertically using Flexbox?
12. What is the difference between display:flex and display:grid? When would you use each?

Unit 2:

1. What is NodeJS and why is it non-blocking?
2. What is the difference between NodeJS and a browser JavaScript environment?
3. What is package.json and what does it contain?
4. What is the difference between dependencies and devDependencies in NPM?
5. What is middleware in Express?
6. What is the difference between app.get() and app.use() in Express?
7. What is async/await in JavaScript?
8. What is the difference between require() and import in NodeJS?
9. What is REST API?
10. What does express.json() middleware do?
11. What is nodemon and why is it used?
12. What is the role of package-lock.json?
13. What is the difference between process.env and config files?
14. What is CORS and how do you enable it in Express?
15. What does npm init do?

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Unit 3:

1. What is Mongoose and how is it different from the native MongoDB driver?
2. What is a Schema in Mongoose?
3. What is the difference between SQL and NoSQL databases?
4. What is a MongoDB document and collection?
5. What is the difference between findOne() and find() in MongoDB?
6. What are cookies and how do they work?
7. What is the difference between cookies, sessions, and JWT?
8. What is bcrypt and why is it used?
9. What does req.body, req.params, and req.query contain in Express?
10. What is indexing in MongoDB?
11. What is the purpose of .env files in NodeJS?
12. How do you connect NodeJS to MongoDB?
13. What are MongoDB operators? Give examples of \$set, \$gt, \$in.

Unit 4:

1. What is React and why is it used?
2. What is the Virtual DOM and how does React use it?
3. What is the difference between a functional component and a class component?
4. What is Hook in React?
5. What is useState Hook?
6. What is useEffect Hook?
7. What is the difference between props and state?
8. What is JSX?
9. What is the Fetch API?
10. What is lifting state up in React?
11. What is React Router?
12. Write a functional React component that displays "Hello, {name}" using props.
13. What is ReactDOM.render() vs ReactDOM.createRoot()?
14. What is the purpose of useCallback and useMemo Hooks?

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Unit 5:

1. What is Cloud Computing? Name the three service models.
2. What is the difference between horizontal and vertical scaling?
3. What is a Docker image vs a Docker container?
4. What is a Dockerfile?
5. What is docker-compose?
6. Write a basic Dockerfile for a NodeJS application.
7. What are the most used Docker CLI commands?
8. What is the difference between docker run and docker-compose up?
9. What is Kubernetes (K8s)?
10. What is a Pod in Kubernetes?
11. What is the difference between a VM (Virtual Machine) and a container?
12. What is VPC (Virtual Private Cloud)?
13. What is EC2 in AWS?
14. What is the difference between Kubernetes Deployment and a Service?
15. What are common kubectl commands used in Kubernetes?